Kedar Joshi
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EDUCATION

Manipal University Jaipur, Jaipur, India **May 2026**
Bachelor of Technology in Computer Science with *Distinction* (CGPA 8.88/10)
Relevant Courses: Data Structure and Algorithms, Artificial Intelligence and Soft Computing, Data Science and Machine Learning

PROFESSIONAL EXPERIENCE

Indian Institute of Science Education and Research (IISER), Kolkata, India **May 2025 – Nov 2025**
Research Intern – LLM Evaluation and Robustness

- Developed LogiQAte, a first-of-its-kind diagnostic benchmark with 1,108 questions across four reasoning tasks to evaluate LLMs' robustness under logical obfuscation
- Designed Logifus, a structure-preserving logical obfuscation framework demonstrating that current LLMs exhibit 28.8% average performance degradation under logically equivalent transformations
- Evaluated six state-of-the-art models (GPT-4o, GPT-5, o4-mini, Claude, Gemini, Qwen) revealing significant brittleness in reasoning capabilities
- Conducted layer-wise mechanistic analysis showing 50-80% collapse in next-token confidence under obfuscation, indicating surface-level pattern matching rather than deep reasoning.
- Accepted to EACL 2026 main conference.

India Meteorological Department, Bhopal, India **Nov 2024 – Present**
Research Intern – AI\ML for Atmospheric Sciences

- Develop **Auto-PINN + NAS systems** for automated single-radar 3D atmospheric velocity field reconstruction, demonstrating that physics constraints can serve as information-equivalent substitutes for geometric constraints.
- Implemented physics-informed neural architecture search framework combining DARTS and Bayesian optimization, achieving 200-fold scalability across computational platforms (245K to 50M+ parameters)
- Filed patent application for "Automated Physics-Informed Neural Architecture Search System and Method for Single-Radar Three-Dimensional Atmospheric Velocity Field Reconstruction"

ACADEMIC PROJECTS AND PAPERS

Integrating GLCM Texture Analysis for Improved Deepfake Detection

- Published at IEEE conference | Co-author: Aditya Sinha
- Implemented Grey Level Co-occurrence Matrix (GLCM) texture analysis for deepfake detection on CelebDF(v2) dataset, achieving significant improvements in detection accuracy
- Developed novel approach using Contrast and Dissimilarity features to identify texture inconsistencies in deepfake images
- Publication: https://ieeexplore.ieee.org/document/10939531/authors

Perturbation-Aware Fine-Tuning (PAFT) for Reducing LLM Memorization

- Proposed Perturbation-Aware Fine-Tuning (PAFT), a training-time data augmentation method reducing memorization in LLMs without sacrificing reasoning performance
- Introduced MemGenTOI metric quantifying memorization-generalization trade-off using perplexity score, adversarial compression ratio and test accuracy
- Demonstrated consistent memorization reduction across Qwen2.5 7B, LLaMA 3.1 8B and Mixtral 8x7B models on GSM8K dataset
- Released GSM8K-P dataset with 5,814 perturbation-validated samples for privacy-safe model training.

TECHNICAL SKILLS

- **Programming languages:** Python, C
- **Deep Learning Frameworks:** Pytorch, Tensorflow
- **ML/AI Libraries:** NumPy, Pandas, scikit-learn, OpenCV, scikit-image, Matplotlib, Unsloth
- **Specialized Skills:** Physics-Informed Neural Networks (PINNs), Neural Architecture Search (NAS), Computer Vision, Natural Language Processing, LLM Evaluation, Benchmark Development
- **Tools and Technologies:** CUDA, OpenAI, Prover9, Git

COURSES & CERTIFICATIONS

- **Generative AI with Large Language Models** | DeepLearning.AI & AWS | Feb 2025
- **Prediction and Control with Function Approximation** | University of Alberta | Dec 2024
- **Sample-based Learning Methods** | University of Alberta | Nov 2024
- **Fundamentals of Reinforcement Learning** | University of Alberta | Nov 2024